PRODUCT USER MANUAL

For Baltic Sea Sea Surface Temperature L3S Reprocessed Product

SST_BAL_PHY_L3S_MY_010_040

Issue: 4.0

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2013/08/19	All	Creation of the document.	Jacob L. Høyer	L. Crosnier
1.1	**2015/05/01**	All	Change format to fit CMEMS graphical rules		L. Crosnier
1.2	**2016/01/15**	All	Rebranded for CMEMS	Jacob L. Høyer	B. Hackett
1.3	**2017/09/07**	All	Updated to include 2010+2011 results	Jacob L. Høyer	
1.4	**2020/09/09**	All	Upstream data change, temporal extension and new template	Jacob L. Høyer	
1.5	**2021/09/10**	All	Regular time extension	Wiebke Kolbe, Ioanna Karagali, Jacob L. Høyer	
2.0	**2022/06/15**	All	Creation of the document in new template, inclusion of information about the new L3S product	Ioanna Karagali	Copernicus Marine Product management
3.0	**2023/06/15**	I.2, II.3	Updated for EIS Nov 2023	Jacob L. Høyer, Matilde B. Kreiner	Copernicus Marine Product management
4.0	**2025/06/15**	All	Full Reprocessing using ESA CCI SST v3.0 for the period 1982-2021, ICDR SST for 2022-2024, New Template	Ioanna Karagali	Copernicus Marine Product management

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis).

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the Product User Manual describing the Copernicus Marine Service SST reprocessed (level 3S) product over the Baltic Sea and North Sea (SST_BAL_PHY_L3S_MY_010_040), processed at the Danish Meteorological Institute (DMI). The manual covers: how the product is created, content, and which data services are available to access it.
The Copernicus Marine Baltic Sea SST reprocessed L3S product is the multi-sensor super-collated product where all satellite SST observations have been aggregated but where no interpolation has been performed. It is created by created by the DMIOI system (Høyer and She, 2007; Høyer and Karagali, 2016) using L2P data from the re-processed ESA SST CCI SST v3.0 (Embury et al., 2024) for the period 1982-2021, L3C SST from C3S SST ICDR for 2022, L3C SST ICDR funded by the UK Earth Observation Climate Information Service (EOCIS) and Marine and Climate Advisory Service (MCAS) and high-resolution sea ice products from the Copernicus Marine Service (SEAICE_BAL_PHY_L4_MY_011_019).
Due to changes in the upstream data format, the L3S product may seem different when plotting from 2022 to July 19th 2024, as the input data from the ESA CCI SST v3.0 on L2P format are available only until end of 2021 and from 2022 onwards, the L3U SST ICDR data from the C3S EOCIS/MCAS projects are used; these are already gridded on a coarser resolution grid than that of the Copernicus Marine Baltic Sea SST reprocessed L3S product.
From 20th July 2024, the upstream SST input data have been made publicly available on L2P format so the L3S product will appear as it was up to end of 2021 from 20th July 2024 onward.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	SST_BAL_PHY_L3S_MY_010_040
Geographical coverage	North Sea & Baltic Sea : 10°W-30°E ; 48°S – 66°N
Variables	• Sea Surface Temperature • SSES Bias • Mask • SSES Standard Deviation • Sea Ice Fraction • Quality Level • Source of SST • Sum SST • Sum Square SST • SST dtime
Product Type	Observations, Multi-Year (MY)
Available time series	1982-2024
Temporal resolution	Daily
Update Frequency	T-3 months
Target delivery time	N/A
Horizontal resolution	0.02 degrees
Number of vertical levels	Surface only
Format	NetCDF4, CF1.4 and GHRSST GDS2.1

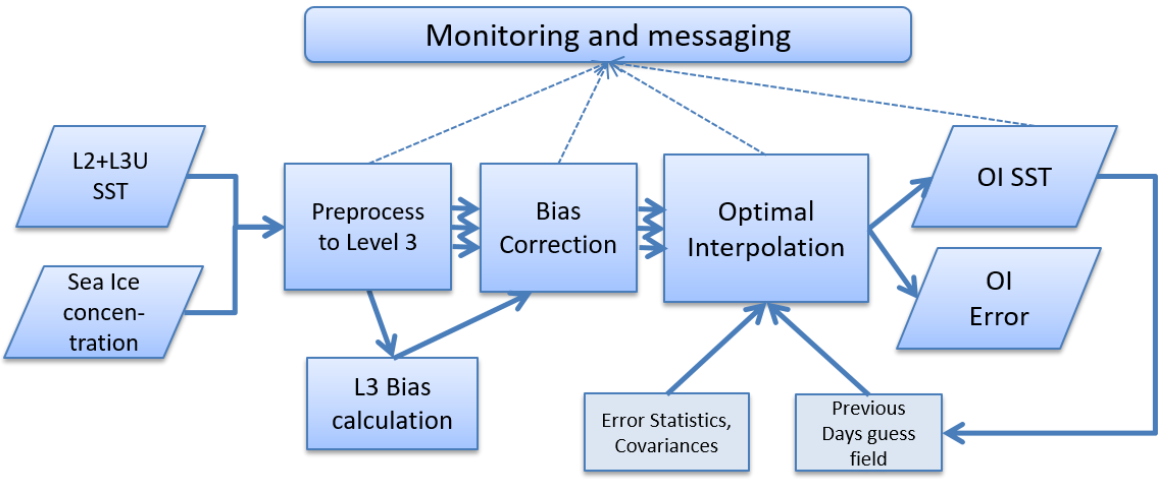
b) Production System Description

All satellite data valid for a particular day, within 24 hours from the analysis are considered. The input L2/L3U SST data undergo various QC and processing steps to generate separate level 3 products:

- Only satellite SST satellite data, which have a quality flag of 4 and higher are accepted
- The SST CCI and C3S data have been adjusted to 20 cm, which reduces the contamination from diurnal warming but to further minimize these effects, observations between 10 and 18 local time have not been used.

The main SST analysis uses a persistence approach based on the use of the previous analysis field as a first guess. The collated L3 (L3S) SST observations from the current day are interpreted as anomalies with respect to the first guess field and local statistics have been included to estimate the optimal estimates SST values for the final field (Høyer and She, 2007). The ESA CCI SST project performed adjustments of the different satellite products against each other and no bias corrections have therefore been applied in this reprocessing.

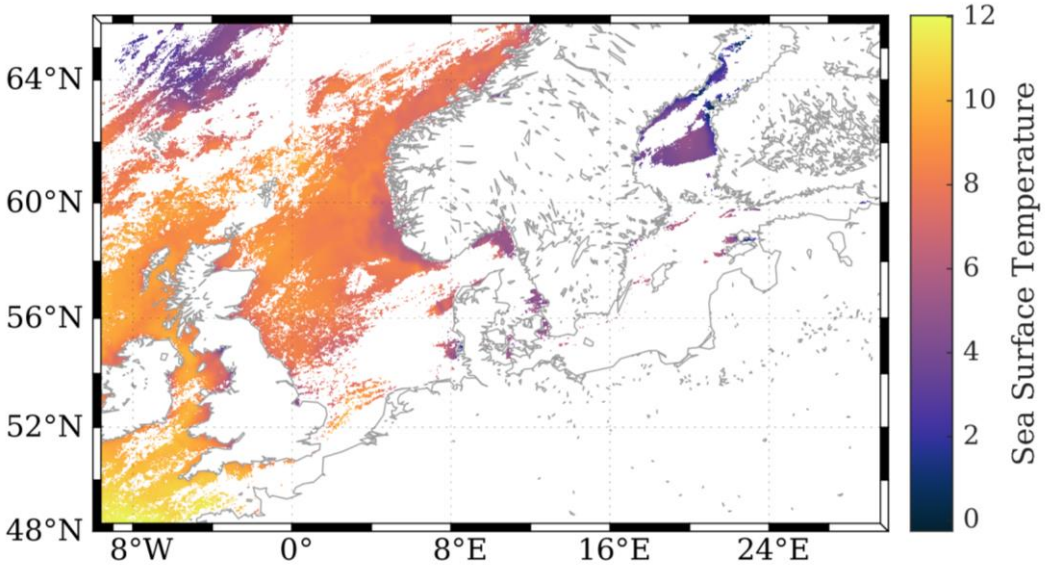
A high resolution land mask is used, which is a modified version of the Globolakes dataset (Carrea et al., 2015; GloboLakes: high-resolution global limnology dataset v1) obtained through the ESA SST_cci project (Globolakes-static_distance_to_land_Map-300m-P5Y-2005-ESACCI_WB-fv1.0_RES120_125km.nc) after personal communication with the Project Lead.



The reprocessing system running at DMI.

North Sea and Baltic Sea (10°W-30°E; 48°S – 66°N): 1/50°, regular grid, 901 x 2001

This product is regional with dedicated projection and spatial resolution. It is defined on a standard collocated grid at 1/50 degree (approx. 2 km).



Vertical grid	Surface
System Name	NSB DMIOI
Model version	
Horizontal Resolution	1/50°
Vertical Resolution	Surface
Interpolation method	N/A
Coupling	N/A
Sea Ice Model	N/A
Bathymetry	N/A
Initial conditions	N/A
Atmospheric forcings	N/A
Tidal constituents	N/A
Rivers	N/A
Data Assimilation scheme	N/A
Assimilated observations	SST satellite data: A collection of Infrared L2P/L3U data from ESA SST CCI and C3S projects: • ESA CCI SST CDR v3.0 1982-2021 • C3S L3U ICDR 2022 • L3U ICDR 2023-Jul 2024 (funded by UK EOCIS/MCAS)

	• L2P ICDR Jul 2024- (funded by UK EOCIS/MCAS) See quality information document for full details. Sea-ice concentration data: Baltic high resolution sea ice concentration data from the Copernicus Marine Service SI TAC (SEAICE_BAL_PHY_L4_MY_011_019).

c) Dataset details

Product Lines	SST_BAL_PHY_L3S_MY_010_040	
Geographical coverage	North Sea and Baltic Sea	
Datasets: cmems_obs-sst_bal_phy_my_l3s_P1D-m		
Product Type	Analysis	
Update frequency	T-3 months	
Available time series	1982up to T-3months	
Target delivery time	T-3 months	
Temporal resolution	Daily	

Dataset : cmems_obs-sst_bal_phy_my_l3s_P1D-m contains the 2D daily subskin SST fields
sea_surface_temperature [K] **sea surface foundation temperature** **sea_surface_temperature**
sses_standard_deviation [K] SSES standard deviation
sses_bias [K] **SSES bias error**
quality_level quality level of SST pixel
source_of_sst SST product origin
sum_sst [C] sum of contributing pixel sst values
sum_square_sst [C2] sum of contributing pixel sst value squares
sst_dtime [seconds] time difference from reference time
or_number_of_sst_pixels number of pixels from the L2Ps contributing to the SST value
sea_ice_fraction sea ice area fraction sea_ice_area_fraction
mask land sea ice lake bit mask

d) Processing information

Product disseminating schedule :

Type of product	Reference Time	Product temporal coverage (hours)	Product disseminating scheduled time	Dissemination day
Analysis	00 UTC	0 to 24	T-3 months	Last weekday of month

e) Catalogue updating

The catalogue data from 1982 to T-6months before present are not updated anymore.

The product is updated **regularly, at T-3 months intervals**, by **1200 UTC of the last weekday of the month**.

f) Time averaging

Dataset frequency	Time averaging	Forecast lead time
	Each daily SST analysis is based on aggregated input data from the analysis day.	N/A

g) Product use and impact

For applications like regional, ocean and atmospheric modelling, where assimilation of SST observations is needed there are several reasons to choose the L3S product specifically developed for the Baltic Sea and North Sea, instead of one of the global products. Some of the advantages are listed below:

- Focus on optimal merging of all available satellite SSTs throughout the record (1982-2024 currently).
- Increased spatial resolution gives a better representation of fronts and near coastal effects.
- Regional, high resolution land mask gives a better representation of small islands and complex coastal regions.
- The high resolution 5 km sea ice product derived from ice charts is significantly better in this coastal region than the global Microwave-based products.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

4.1) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
cmems_obs-sst_bal_phy_my_l3s_P1D-m	<date>000000-DMI-L3S_GHRSST-SSTsubskin-night_SST_UHR_MY-NSEABALTIC-v02.0-fv01.0.nc	0.01

4.2) STRUCTURE OF OF FILES

Examples of the header of output NetCDF files are inserted in **annex,** for each dataset.

5) REFERENCES

Quality Information Document
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SST-QUID-010-016-040.pdf

Høyer, J. L. and She, J. (2007) Optimal interpolation of sea surface temperature for the North Sea and Baltic Sea, J. Mar. Sys., Vol 65, 1-4, pp.

Høyer, J. L., & Karagali, I. (2016). Sea surface temperature climate data record for the North Sea and Baltic Sea. Journal of Climate, 29(7), 2529-2541.

Embury, O., Merchant, C.J., Good, S.A., Rayner, N.A., Høyer, J.L., Atkinson, C., Block, T., Alerskans, E., Pearson, K.J., Worsfold, M., McCarroll, N., Donlon, C. Satellite-based time-series of sea-surface temperature since 1980 for climate applications. Scientific Data 11, 326 (2024). https://doi.org/10.1038/s41597-024-03147-w

Baltic Sea ice concentration, extent, and classification time series (SEAICE_BAL_PHY_L4_MY_011_019). E.U. Copernicus Marine Service Information (CMEMS). Marine Data Store (MDS). DOI: 10.48670/moi-00131 (Accessed on 30-Jan-2025)

The Recommended GHRSST-PP Data Specification GDS Version 2.1, https://doi.org/10.5281/zenodo.6984989

Carrea, L., Embury, O., and Merchant, C.J. (2015), GloboLakes: high-resolution global limnology dataset v1. Centre for Environmental Data Analysis, 21 July 2015. doi:10.5285/6be871bc-9572-4345-bb9a-2c42d9d85ceb. https://dx.doi.org/10.5285/6be871bc-9572-4345-bb9a-2c42d9d85ceb

6) ANNEX

a) For dataset cmems_obs-sst_bal_phy_my_l3s_P1D-m

```
netcdf \20230105000000-DMI-L3S_GHRSST-SSTsubskin-night_SST_UHR_MY-NSEABALTIC-v02.0-fv01.0 {

dimensions:

        lon = 2001 ;

        lat = 901 ;

        time = 1 ;

variables:

        float lon(lon) ;

                lon:standard_name = "longitude" ;

                lon:long_name = "longitude" ;

                lon:units = "degrees_east" ;

                lon:axis = "X" ;

                lon:valid_min = -10.f ;

                lon:valid_max = 30.f ;

                lon:comment = "geographical coordinates, WGS84 projection" ;

        float lat(lat) ;

                lat:standard_name = "latitude" ;

                lat:long_name = "latitude" ;

                lat:units = "degrees_north" ;

                lat:axis = "Y" ;

                lat:valid_min = 48.f ;

                lat:valid_max = 66.f ;

                lat:comment = "geographical coordinates, WGS84 projection" ;

        int time(time) ;

                time:standard_name = "time" ;

                time:long_name = "reference time of sst field" ;
```

time:units = "seconds since 1981-01-01 00:00:00" ;

time:axis = "T" ;

time:comment = "Nominal time of Level 3" ;

short sea_surface_temperature(time, lat, lon) ;

sea_surface_temperature:standard_name = "sea_surface_temperature" ;

sea_surface_temperature:long_name = "sea surface foundation temperature" ;

sea_surface_temperature:units = "kelvin" ;

sea_surface_temperature:scale_factor = 0.01f ;

sea_surface_temperature:add_offset = 273.15f ;

sea_surface_temperature:_FillValue = -32768s ;

sea_surface_temperature:comment = "Foundation sea surface temperature of the ocean" ;

sea_surface_temperature:source = "OSI-SAF and GHRSST L2P satellite SST products" ;

sea_surface_temperature:valid_min = -300s ;

sea_surface_temperature:valid_max = 4500s ;

byte sses_standard_deviation(time, lat, lon) ;

sses_standard_deviation:long_name = "SSES standard deviation" ;

sses_standard_deviation:units = "kelvin" ;

sses_standard_deviation:scale_factor = 0.02f ;

sses_standard_deviation:add_offset = 2.54f ;

sses_standard_deviation:_FillValue = -128b ;

sses_standard_deviation:comment = "Standard deviation estimate" ;

sses_standard_deviation:valid_min = -127b ;

sses_standard_deviation:valid_max = 127b ;

byte sses_bias(time, lat, lon) ;

sses_bias:long_name = "SSES bias error" ;

sses_bias:units = "kelvin" ;

sses_bias:scale_factor = 0.01f ;

```
            sses_bias:add_offset = 0.f ;
            sses_bias:_FillValue = -128b ;
            sses_bias:comment = "SSES bias error" ;
            sses_bias:valid_min = -127b ;
            sses_bias:valid_max = 127b ;
        byte mask(time, lat, lon) ;
            mask:long_name = "land sea ice lake bit mask" ;
            mask:_FillValue = -128b ;
            mask:comment = "Mask can be used to further filter the data" ;
            mask:flag_meanings = "water land optional_lake_surface sea_ice
optional_river_surface" ;
            mask:source = "NAVOCEANO_landmask_v1.0, Myocean_HR_ice" ;
            mask:flag_masks = 1b, 2b, 4b, 8b, 16b ;
            mask:valid_min = 0b ;
            mask:valid_max = 31b ;
        byte sea_ice_fraction(time, lat, lon) ;
            sea_ice_fraction:standard_name = "sea_ice_area_fraction" ;
            sea_ice_fraction:long_name = "sea ice area fraction" ;
            sea_ice_fraction:units = "1" ;
            sea_ice_fraction:scale_factor = 0.01f ;
            sea_ice_fraction:add_offset = 0.f ;
            sea_ice_fraction:_FillValue = -128b ;
            sea_ice_fraction:comment = "This will be different for each system" ;
            sea_ice_fraction:source = "MyOcean HR icemask" ;
            sea_ice_fraction:valid_min = 0b ;
            sea_ice_fraction:valid_max = 100b ;
        byte quality_level(time, lat, lon) ;
            quality_level:long_name = "quality level of SST pixel" ;
```

quality_level:_FillValue = -128b ;

quality_level:comment = "These are the overall quality indicators and are used for all GHRSST SSTs" ;

quality_level:flag_values = 0b, 1b, 2b, 3b, 4b, 5b ;

quality_level:flag_meanings = "no_data bad_data worst_quality low_quality acceptable_quality best_quality" ;

quality_level:valid_min = 0b ;

quality_level:valid_max = 5b ;

short source_of_sst(time, lat, lon) ;

source_of_sst:long_name = "SST product origin" ;

source_of_sst:units = "1" ;

source_of_sst:_FillValue = -32768s ;

source_of_sst:comment = "code: EUR-L2P-NARNPP_SST = 1; NAVO-L2P-AVHRR19_L = 2; VIIRS_N20-OSPO-L2P = 4;NAVO-L2P-NPP = 8; EUR-L2P-SLSTR = 16 EUR-L3-SEVIRI_SST = 32;GOES16-OSISAF-L3C = 64; EUR-L2P-AVHRR_METOP_A = 128" ;

source_of_sst:flag_masks = 1s, 2s, 4s, 8s, 16s, 32s, 64s, 128s, 256s, 512s ;

source_of_sst:valid_min = 0s ;

source_of_sst:valid_max = 32767s ;

float sum_sst(time, lat, lon) ;

sum_sst:long_name = "sum of contributing pixel sst values" ;

sum_sst:units = "Celsius" ;

sum_sst:scale_factor = 1.f ;

sum_sst:add_offset = 0.f ;

sum_sst:_FillValue = -1000.f ;

sum_sst:comment = "Sum of original contributing pixel sst values" ;

sum_sst:valid_min = -999.f ;

sum_sst:valid_max = 90000.f ;

float sum_square_sst(time, lat, lon) ;

sum_square_sst:long_name = "sum of contributing pixel sst value squares" ;

sum_square_sst:units = "Celsius2" ;

sum_square_sst:scale_factor = 1.f ;

sum_square_sst:add_offset = 0.f ;

sum_square_sst:_FillValue = -1.f ;

sum_square_sst:comment = "Sum of original contributing pixel sst value squares" ;

sum_square_sst:valid_min = 0.f ;

sum_square_sst:valid_max = 900000.f ;

int sst_dtime(time, lat, lon) ;

sst_dtime:long_name = "time difference from reference time" ;

sst_dtime:units = "seconds" ;

sst_dtime:_FillValue = -2147483648 ;

sst_dtime:comment = "time plus sst_dtime gives seconds after 00:00:00 UTC January 1, 1981" ;

sst_dtime:valid_min = -2147483647 ;

sst_dtime:valid_max = 2147483647 ;

short or_number_of_sst_pixels(time, lat, lon) ;

or_number_of_sst_pixels:long_name = "number of pixels from the L2Ps contributing to the SST value" ;

or_number_of_sst_pixels:_FillValue = -32768s ;

or_number_of_sst_pixels:comment = "Original number of pixels from the L2Ps contributing to the SST value" ;

or_number_of_sst_pixels:valid_min = 0s ;

or_number_of_sst_pixels:valid_max = 32767s ;

// global attributes:

:Conventions = "CF-1.6" ;

:title = "North Sea/Baltic Sea SST, L3S, (SST_BAL_PHY_L3S_MY_010_040)" ;

:summary = "DMI Sea Surface Temperature supercollated level 3 field (L3S) using multisensor night-time satellite SST observations" ;

:references = "Høyer, J. L., Le Borgne, P., & Eastwood, S. (2014). A bias correction method for Arctic satellite sea surface temperature observations. Remote Sensing of Environment, 146, 201-213." ;

:institution = "Danish Meteorological Institute, DMI" ;

:history = "Version 1.0" ;

:license = "these data are available free of charge under the Copernicus CMEMS data policy" ;

:id = "DMI-L3S_GHRSST-SSTsubskin-night_SST_UHR_MY-NSEABALTIC" ;

:naming_authority = "org.ghrsst" ;

:product_version = "Version 1.0" ;

:uuid = "eaa83c4b-40fa-49dd-b8b2-4ae3a1e87096" ;

:gds_version_id = "2.0" ;

:netcdf_version_id = "netcdf-4.3.3.1, 27 February 2015" ;

:date_created = "20241216T025414Z" ;

:file_quality_level = 3 ;

:spatial_resolution = " 0.02 degrees" ;

:start_time = "20230105T000000Z" ;

:time_coverage_start = "20230105T000000Z" ;

:stop_time = "20230106T000000Z" ;

:time_coverage_end = "20230106T000000Z" ;

:westernmost_longitude = -10.f ;

:easternmost_longitude = 30.f ;

:southernmost_latitude = 48.f ;

:northernmost_latitude = 66.f ;

:source = "ESA SST CCI v3.0, C3S and CMEMS FMI and SMHI Sea ice concentration" ;

:platform = "NOAA, ERS, ENVISAT, SENTINEL 3, MetOp" ;

:sensor = "AVHRR, (A)ATSR, SLSTR" ;

:Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;

:metadata_link = "http://data.nodc.noaa.gov/NESDIS_DataCenters/metadata/index.html" ;

:keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;

:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;

:standard_name_vocabulary = "NetCDF Climate and forecast (CF) Metadata Convention" ;

:geospatial_lat_units = "degrees north" ;

:geospatial_lon_units = "degrees east" ;

:geospatial_lat_resolution = 0.02f ;

:geospatial_lon_resolution = 0.02f ;

:acknowledgment = "Please acknowledge the use of these data with the following statement: These data were provided by DMI and Copernicus CMEMS" ;

:creator_name = "Danish Meteorological Institute - DMI Copenhagen" ;

:creator_email = "myocean_po@dmi.dk" ;

:creator_url = "http://ocean.dmi.dk/satellite/" ;

:project = "Copernicus Marine Environmental Monitoring Service (CMEMS) SST-TAC" ;

:publisher_name = "CMEMS SST TAC and DMI" ;

:publisher_url = "https://marine.copernicus.eu/" ;

:publisher_email = "servicedesk.cmems@mercator-ocean.eu" ;

:processing_level = "L3S" ;

:cdm_data_type = "grid" ;

:comment = "IN NO EVENT SHALL DMI OR ITS REPRESENTATIVES BE LIABLE FOR ANY DAMAGES WHATSOEVER INCLUDING, WITHOUT LIMITATION, SPECIAL, INDIRECT, INCIDENTAL OR CONSEQUENTIAL DAMAGES OR DAMAGES FOR LOSS OF BUSINESS PROFITS OR SAVINGS, BUSINESS INTERRUPTION, LOSS OF BUSINESS INFORMATION OR OTHER PECUNIARY LOSS ARISING OUT OF THE USE OF OR THE BUSINESS INTERRUPTION, LOSS OF BUSINESS INFORMATION OR OTHER PECUNIARY LOSS ARISING OUT OF THE USE OF OR THE INABILITY TO USE THIS DMI PRODUCT, EVEN IF DMI HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. THIS LIMITATION SHALL APPLY TO CLAIMS OF PERSONAL INJURY TO THE EXTENT PERMITTED BY LAW. SOME COUNTRIES OR STATES DO NOT ALLOW THE EXCLUSION OR LIMITATION OF LIABILITY FOR CONSEQUENTIAL, SPECIAL, INDIRECT, INCIDENTAL DAMAGES AND, ACCORDINGLY, SOME PORTIONS OF THESE LIMITATIONS MAY NOT APPLY TO YOU. BY USING THIS PRODUCT, YOU HAVE ACCEPTED THAT THE ABOVE LIMITATIONS OR THE MAXIMUM LEGALLY APPLICABLE

SUBSET OF THESE LIMITATIONS APPLY TO YOUR USE OF THIS PRODUCT. WARNING Some applications are unable to properly handle signed bytevalues. If values are encountered > 127, please subtract 256 from this reported value" ;

}